Product Requirements Document: Manufacturing Management System

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This system replaces notebooks, WhatsApp and spreadsheets with one application that tracks every order from quotation to final payment, including its production, material usage, cost and profit.

1. Overview

Problem. Orders, designs, payments, production progress and stock are tracked in separate places (paper, chat, spreadsheets). The owner cannot say, for any order, what it cost to make, who worked on it, how much material was wasted, or whether it made a profit.

Product. A Windows desktop application that runs on one office computer, works fully offline, and keeps all data in a local, protected database. It has five connected modules: Sales, Production, Stock & Purchase, Delivery & Feedback, and Accounts & Reports. Every record links back to an order, so cost and profit are calculated automatically.

Goals

- One record per order, from quotation to final payment
- Production visibility: stage, worker, time taken, quantity completed, waste
- Stock that updates itself when material is used, with low-stock alerts before it runs out
- True production cost and profit margin per order
- Daily and monthly reports without manual calculation

Success metrics

Metric	Target (to be confirmed with client)
Orders with a complete job card	100% within 3 months of go-live
Orders with cost and margin calculated automatically	100%
Stock-outs on tracked materials	Reduced to near zero after alerts go live

Metric	Target (to be confirmed with client)
Time to produce the monthly sales report	Under 1 minute
Daily active use by sales, production and store staff	90% of staff

2. Users and roles

Access is role-based. Each user sees only the screens and figures their role needs; cost, profit and accounts stay restricted to Owner and Accountant.

Role	Main work in the system	Restricted from
Owner / Admin	Full access, approvals, all reports, user management	Nothing
Sales executive	Customers, quotations, orders, invoices, design approval, payment status	Cost, profit, purchase prices
Designer	Upload designs, revise, track customer approval	Pricing, accounts
Production supervisor	Create and assign job cards, track stages, record waste	Accounts, profit
Production worker	See assigned jobs, start and stop a stage, enter quantity done and waste	Everything else
Store / purchase officer	Stock, purchase orders, suppliers, goods receipt	Sales prices, profit
Delivery staff	Delivery list, delivery note, proof of delivery	Cost, profit
Accountant	Payments, expenses, pending dues, cost and profit reports	Production stage editing

3. Scope

Version 1 covers the five modules the client listed. Anything not named here is out of scope until agreed.

In scope

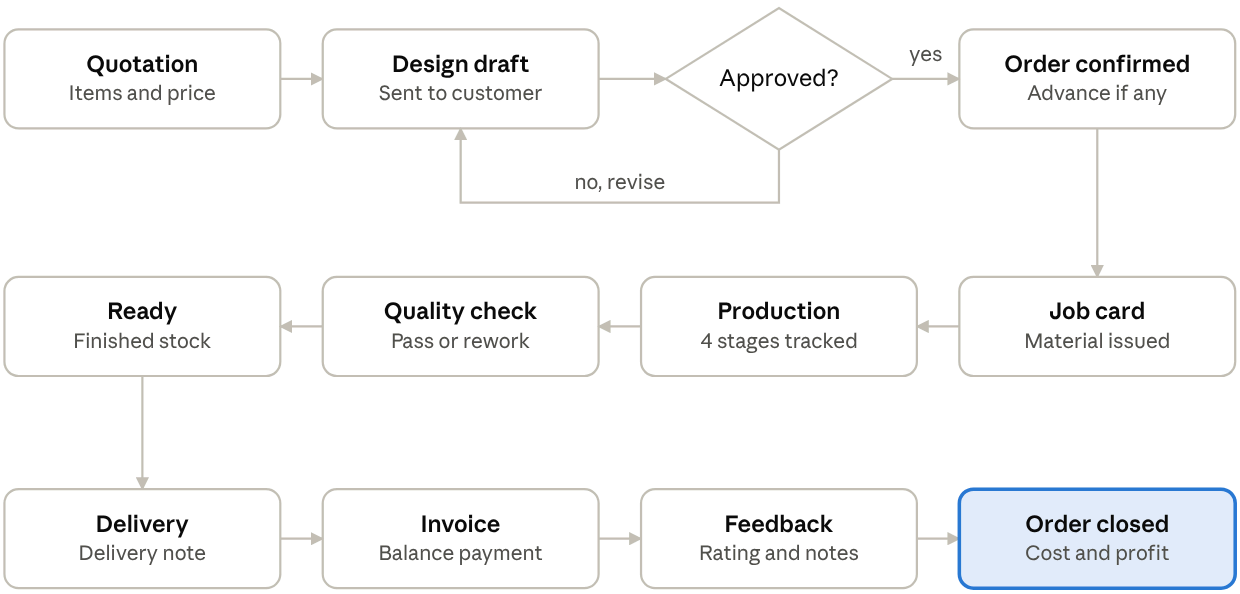
- Sales, production, stock and purchase, delivery, accounts and reports, as specified in sections 5 to 9
- Windows desktop application installed on one computer, usable without internet
- Malayalam and English interface and printed documents
- Role-based login and an audit trail on key changes
- PDF output for quotations, invoices, purchase orders and delivery notes
- Excel/PDF export of every report

Out of scope for v1

- Full double-entry accounting, GST filing and payroll
- Online customer portal or e-commerce storefront
- Machine integration (laser, printer) and barcode hardware
- Native mobile apps, a web version, and use from several computers over a network
- Automated WhatsApp/SMS sending (PDFs can be saved and shared manually in v1; see open questions)

4. Order lifecycle

Every order follows one chain and ends with its cost and profit



order lifecycle · 12 steps, 1 decision

A customer design must be approved before the order is confirmed. The job card is created at confirmation, and the order closes only after delivery, balance payment and feedback, when its production cost and profit are fixed.

5. Module 1: Sales Management

Priority key: **M** = must have for v1, **S** = should have, **C** = could have.

ID	Requirement	Priority
SM-01	Customer master: name, company, phone, alternate phone, email, billing and delivery address, GST number (optional), customer type (retail / wholesale), notes	M
SM-02	Search customers by name or phone; warn on duplicate phone number	M
SM-03	Customer history: all quotations, orders, invoices, payments and outstanding balance on one page	M
SM-04	Quotation: select customer, add line items (product, size, material, quantity, rate, discount, tax), validity date, terms; auto numbering	M
SM-05	Quotation status: Draft, Sent, Accepted, Rejected, Expired; convert an accepted quotation to an order in one click	M
SM-06	Retail and wholesale price lists; wholesale slab pricing by quantity; manual price override with reason, restricted by role	M
SM-07	Order: order number, customer, items, due date, priority, special instructions, attached files	M
SM-08	Design and approval: upload design files and mockups per order, keep versions, record customer approval (who, when, how) before production can start	M
SM-09	Order status: Quotation, Design, Approved, In production, Ready, Delivered, Closed, Cancelled	M

ID	Requirement	Priority
SM-10	Payment status per order: Unpaid, Advance paid, Partially paid, Paid; record each payment with date, amount, mode (cash, UPI, bank, cheque) and reference	M
SM-11	Invoice generated from an order: invoice number, tax breakup, PDF and print; support partial and final invoices	M
SM-12	Credit note or adjustment for returns and corrections, with approval	S
SM-13	Wholesale accounts: credit limit, credit period, overdue highlighting	S
SM-14	Payment reminders list: customers with dues sorted by age of balance	S
SM-15	Order cancellation with reason; block cancellation after production starts unless Owner approves	S

Business rules

- Production cannot start until the design is approved and the order is confirmed.
- A wholesale customer is assigned the wholesale price list by default.
- An invoice cannot be edited after issue; corrections go through a credit note.

6. Module 2: Production Management

Every confirmed order gets one **Production Job Card**. The job card is the single source for what is made, by whom, how long it took and how much material was wasted.

ID	Requirement	Priority
PM-01	Job card created automatically from a confirmed, design-approved order; one order may have several job cards (per product or batch)	M

ID	Requirement	Priority
PM-02	Job card content: order and customer reference, product, quantity, specifications, approved design file, due date, priority, special notes	M
PM-03	Configurable stages per product type. Default stages: Cutting, Printing, Assembly, Finishing; admin can add, rename or reorder stages (for example Stitching, Packing)	M
PM-04	Assign each stage to a worker or team; reassignment is logged	M
PM-05	Stage statuses: Not started, In progress, On hold, Completed; a stage cannot start before the previous stage is completed unless the supervisor overrides	M
PM-06	Worker screen (large buttons, touchscreen friendly): list of assigned jobs, Start and Stop buttons per stage, quantity completed	M
PM-07	Quantity tracking per stage: planned, completed, rejected; progress percentage for the job	M
PM-08	Time tracking per stage: start time, end time, pauses; total time taken; manual correction by supervisor with reason	M
PM-09	Material issue: raw materials planned and actually issued to the job from stock	M
PM-10	Waste entry per stage: material, quantity, reason (cutting offcut, machine error, design error, damage); waste value calculated from material cost	M
PM-11	Reusable offcut recorded back to stock as scrap or remnant, with size where relevant	S
PM-12	Rework entry when a stage fails quality check; rework time and material counted against the job	S
PM-13	Production board: all jobs by stage and due date, overdue jobs highlighted	M

ID	Requirement	Priority
PM-14	Printable job card with order details, design thumbnail and checklist	S
PM-15	Worker productivity report: jobs, quantity, time and waste per worker and per stage	S
PM-16	Labour cost per job from time taken and worker hourly or piece rate	S

Business rules

- Finishing completion moves the order to Ready and adds finished goods to stock (if made for stock) or reserves them for the order.
- Material issued minus material used minus recorded waste must reconcile; variances are flagged to the supervisor.

7. Module 3: Stock and Purchase Management

ID	Requirement	Priority
ST-01	Raw material master: name, category, unit of measure, specification (for example thickness, size, colour), default supplier, standard cost	M
ST-02	Raw material stock: current quantity, reserved quantity and available quantity per item; optional multiple storage locations	M
ST-03	Finished product master and finished goods stock: product, variant, quantity on hand, quantity reserved for orders	M
ST-04	Every stock change is a ledger entry: opening, purchase receipt, issue to job, return, waste, adjustment, with date, user and reference	M
ST-05	Automatic deduction of raw material when it is issued to a job card	M

ID	Requirement	Priority
ST-06	Material usage and wastage report per job, per material and per period; waste percentage against issued quantity	M
ST-07	Reorder level and minimum stock per material; low-stock alert on the dashboard, and by email or notification, before stock reaches zero	M
ST-08	Stock forecast hint: days of stock left based on recent average usage	C
ST-09	Supplier master: name, contact person, phone, address, GST number, materials supplied, payment terms, notes	M
ST-10	Purchase order: supplier, items, quantity, rate, tax, expected date, terms; statuses Draft, Approved, Sent, Partially received, Received, Cancelled	M
ST-11	Goods receipt against a purchase order: received quantity, rejected quantity, supplier bill number; stock updates on receipt	M
ST-12	Purchase price history per material and supplier; weighted average cost used in job costing	S
ST-13	Supplier payments: bills, amount paid, balance payable, due dates	S
ST-14	Physical stock count with variance adjustment, requiring Owner approval	S
ST-15	Suggested purchase list: materials below reorder level, grouped by supplier, convertible to purchase orders	S

Business rules

- Stock cannot go negative unless an Owner override is recorded.
- Material cost on a job is taken from the weighted average cost at the time of issue.
- Low-stock alerts repeat daily until a purchase order is raised for the item.

8. Module 4: Delivery, Delivery Note and Feedback

ID	Requirement	Priority
DL-01	Delivery created from a Ready order; supports full and partial deliveries against one order	M
DL-02	Delivery details: delivery date, mode (own vehicle, courier, customer pickup), vehicle or tracking number, delivery person, address	M
DL-03	Delivery note (PDF and print): delivery note number, customer, order reference, items and quantities, signature area; no prices on the note unless chosen	M
DL-04	Delivery statuses: Pending, Out for delivery, Delivered, Failed or returned; reason required for failure	M
DL-05	Proof of delivery: receiver name, signature or photo upload, date and time	S
DL-06	Delivery list for the day, grouped by area, for the delivery staff	S
DL-07	Delivery charge recorded against the order and included in order cost	S
DL-08	Order moves to Delivered automatically when all items are delivered; balance payment reminder shown if dues remain	M
DL-09	Feedback capture per delivered order: star rating (1 to 5), comment, quality, delivery and service ratings, entered by staff	M
DL-10	Complaint record linked to an order: issue, photo, status, resolution (reprint, refund, credit note)	S
DL-11	Feedback report: average rating by period, product and worker group; list of low ratings for follow-up	S

9. Module 5: Accounts and Reports

Accounts

ID	Requirement	Priority
AC-01	Income (receipts): every customer payment linked to an order and invoice	M
AC-02	Expense entry: category (raw material, salary, rent, electricity, transport, maintenance, other), amount, date, mode, supplier or payee, attachment	M
AC-03	Pending payments: customer receivables by age (0 to 30, 31 to 60, 61 to 90, over 90 days) and supplier payables	M
AC-04	Cash and bank book: daily balance by account (cash, bank, UPI)	S
AC-05	Order costing: for each order, material cost, waste cost, labour cost, delivery cost and allocated overhead, summed into production cost	M
AC-06	Profit and margin per order, product, customer and period	M
AC-07	Overhead allocation method set by Owner (for example percentage of material cost, or rate per machine hour)	S
AC-08	Export of ledger-style data to Excel for the client's accountant; accounting software integration is a later phase	S

The margin shown for each order uses this definition, so every report agrees:

$$\text{Production cost} = \text{Material} + \text{Waste} + \text{Labour} + \text{Delivery} + \text{Overhead}$$
$$\text{Profit} = \text{Invoice value (excluding tax)} - \text{Production cost}$$
$$\text{Margin \%} = \frac{\text{Profit}}{\text{Invoice value (excluding tax)}} \times 100$$

Reports

All reports filter by date range, customer, product and sales type (retail / wholesale), and export to PDF and Excel.

ID	Report	Audience	Priority
RP-01	Daily sales report: orders, invoices, receipts, retail vs wholesale split	Owner, Accountant	M
RP-02	Monthly sales report: totals, trend against previous month, top customers and products	Owner	M
RP-03	Income and expense statement: money in, money out, net	Owner, Accountant	M
RP-04	Pending payments and aged receivables	Owner, Accountant, Sales	M
RP-05	Order profitability: cost breakdown, profit and margin per order	Owner	M
RP-06	Production report: jobs by stage, delays, time per stage, waste	Owner, Supervisor	M
RP-07	Stock report: current stock, low stock, usage and wastage by material	Owner, Store	M
RP-08	Purchase report: purchases by supplier and material, pending purchase orders	Owner, Store	S
RP-09	Delivery and feedback report	Owner	S
RP-10	Owner dashboard: today's sales, receipts, orders due, overdue jobs, low-stock items, dues	Owner	M

10. Non-functional requirements

Area	Requirement
Language	Interface in English and Malayalam, switchable per user; customer-facing documents (quotation, invoice, delivery note) printable in either language; Malayalam text must render correctly in PDFs
Devices	Windows desktop application on one office computer. A simple, touch-friendly worker screen with large buttons for the shop floor, usable on a touchscreen monitor or tablet-style PC
Performance	Screens open in under 2 seconds; reports for one month generate in under 10 seconds
Availability	Works fully offline; no internet needed for daily use. Reliable restart and recovery after power cuts without data loss
Security	Login with role-based permissions; password hashing; auto-lock after inactivity; encrypted local database; no remote access; secure separation between the app screens and the database
Audit trail	Log who created, edited or deleted orders, prices, payments, stock adjustments and invoices, with before and after values
Backup	Automatic daily backup with at least 30 days retention, plus a copy to an external drive; documented restore procedure
Data export	Any list or report exportable to Excel or PDF; full data export available to the Owner
Scalability	Single-computer use with several user accounts; supports at least 50,000 orders without slowing down
Usability	A new staff member can create a quotation or record a stage completion after under 30 minutes of training

Area	Requirement
Data import	One-time import of existing customers, suppliers, materials and opening stock from Excel
Compliance	Invoice format suitable for GST where the client is registered; tax rates configurable

11. Data entities and key links

The order is the hub: sales, production, stock, delivery and accounts all attach to it.

Entity	Key fields	Links to
Customer	name, phone, address, type, credit limit	Quotation, Order
Quotation	number, date, validity, status, total	Customer, Order
Order	number, due date, status, payment status, total	Customer, Design, Job card, Delivery, Invoice
Design	file, version, approval status, approved by, approved on	Order
Invoice and Payment	number, amount, tax, mode, date	Order, Customer
Job card	number, product, quantity, status	Order, Stage log, Material issue
Stage log	stage, worker, start, end, quantity done, rejected	Job card, Worker
Material	name, unit, reorder level, average cost	Stock ledger, Purchase order line
Stock ledger entry	material or product, type, quantity, date, reference	Material, Job card, Purchase receipt
Waste entry	material, quantity, reason, value	Job card, Stage log
Supplier and Purchase order	supplier, items, status, receipt	Material, Stock ledger

Entity	Key fields	Links to
Delivery and Feedback	date, mode, status, receiver, rating	Order
Expense	category, amount, date, mode	Accounts report
User and Role	name, role, permissions, language	All records (audit trail)

Cross-module rules

- Deleting is disabled for orders, invoices, payments and stock entries; records are cancelled or reversed, never removed.
- Material cost, waste cost and labour cost flow from the job card into order costing without re-entry.
- Order totals, payment status and stock balances are always calculated from underlying entries, never typed in.

12. Release plan

Proposed phasing, to be fixed after estimation. Each phase ends with a go-live review with the client before the next begins.

1. **Phase 1: Foundation and Sales.** Users and roles, customers, quotations, orders, design approval, invoices, payments (SM-01 to SM-11), owner dashboard basics. Gate: the sales team runs live orders in the system.
2. **Phase 2: Production.** Job cards, stages, worker screen, time and waste entry (PM-01 to PM-10, PM-13). Gate: every new order has a job card with all stages recorded.
3. **Phase 3: Stock and Purchase.** Materials, stock ledger, automatic deduction, suppliers, purchase orders, low-stock alerts (ST-01 to ST-11). Gate: stock balances match a physical count.
4. **Phase 4: Delivery, Accounts and Reports.** Delivery notes, feedback, expenses, order costing, profit margin, daily and monthly reports (DL, AC, RP items marked M). Gate: the Owner confirms order profit figures against manual calculation for a sample of orders.
5. **Phase 5: Refinement.** All S and C items, performance tuning, extra reports, accounting export, training material.

Data migration and training run alongside phases 1 to 3: import of customers, suppliers, materials and opening stock, and short role-based training sessions before each go-live.

13. Open questions

- [ ] Does the client have a registered GST number, and should invoices follow GST format?
- [ ] Which product types are made, and do all follow the same four stages or do some need Stitching or Packing?
- [ ] Are workers paid monthly, daily or per piece? This decides how labour cost is calculated.
- [ ] How should overheads (rent, electricity, machine cost) be allocated to orders?
- [ ] Is an advance payment mandatory before production starts, or optional?
- [ ] Is material tracked by weight, sheet, piece or area? Are offcuts reused?
- [ ] How many users, and how many locations or branches?
- [ ] Should quotations, invoices and reminders be sent by WhatsApp or SMS automatically in a later phase?
- [ ] Is existing data available in Excel for import?
- [ ] Which computer will run the software, and where will backups be kept (external drive, a second copy elsewhere)?
- [ ] Who approves price overrides, discounts and stock adjustments?